

JACOB E. BARCELONA

Software Engineer · Mobile · Web · Machine Learning

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[portfolio: jacobbarcelona.dev](https://portfolio.jacobbarcelona.dev) | github.com/cob-byte | linkedin.com/in/cob-bytes

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Frontend: React, Next.js, React Native, Expo, TanStack Start, NativeWind, TailwindCSS, HTML, CSS

Backend: Django, FastAPI, Flask, Node.js, Express, REST APIs, JWT / session auth

ML & Computer Vision: TensorFlow, PyTorch, OpenCV, scikit-learn, MediaPipe, Dlib, pandas, NumPy

Databases: PostgreSQL, MySQL, SQLite, Supabase, Firebase

Cloud & Tooling: Git / GitHub, Railway, Render, Netlify, CI/CD, AI-assisted development (Codex, Claude Code), WordPress (custom themes & plugins)

EXPERIENCE

Linoflap Technology Philippines Inc.

2024 – 2026

Software Engineer

Ermita, Manila

- Built and delivered production cross-platform mobile apps in React Native and Expo (iOS and Android) for client distribution, including a field lending & collection app (Kolekta) and a seafarer payroll app for IMS Philippines Maritime Corp.
- Developed RESTful APIs with FastAPI and implemented real-time data synchronization between tablet and mobile clients.
- Developed a contactless QR table-ordering web platform (LinoflapQR) for client deployment using TanStack Start, React 19, and TypeScript.
- Built the IMAI Cart Classifier, a computer-vision product served as a production inference API with OpenCV, TensorFlow, and FastAPI.
- Used AI coding tools (OpenAI Codex, Claude Code) to deliver products on time, automate repetitive tasks, and speed up debugging.

Linoflap Technology Philippines Inc.

2024

Software Developer Intern

Ermita, Manila

- Trained and applied TensorFlow 2 models for image classification, segmentation, and object detection across multiple use cases.
- Integrated MediaPipe hand-tracking and face mesh into a Unity game for gesture-based player control.

PROJECTS

MediQS – Healthcare EMR Portal with Outbreak Forecasting *Django · JavaScript · Bootstrap*

Web platform for Manila health centers to manage patient records and scheduling. Built a structural time-series model on DOH case data (2016–2022) to forecast outbreaks and a heuristic algorithm for appointment prioritization. Deployed on Railway for testing with IT professionals.

IoT Baby Cradle with Computer Vision *Flutter · Python · MTCNN · MediaPipe · Raspberry Pi*

Flutter remote control for a Raspberry Pi-powered cradle. Trained MTCNN for infant face and landmark detection to verify safe sleeping position, and used MediaPipe face mesh and pose detection for automated sleep tracking.

WOS Hub – Whiteout Survival Strategy & Calculator App *React · TypeScript · Express*

Self-initiated full-stack web app built in my spare time for the mobile strategy game Whiteout Survival. Gives players a set of guild strategy tools and in-game calculators to plan resources and decisions, with a React/TypeScript frontend over an Express API.

EDUCATION

Pamantasan ng Lungsod ng Maynila

2020 – 2024

B.S. Information Technology — Magna Cum Laude

Manila

Technological Institute of the Philippines

2018 – 2020

STEM Strand — Graduated with Honors

Manila

ACHIEVEMENTS & ACTIVITIES

- **Category Winner** — Nervos Network Devpost Hackathon, Web3 World Challenge (2022)
- Member, Google Developer Student Clubs – PLM (2020–2024); participant in Hack-Attack Ideathon (2023) and Philippine Web3 Festival Hackathon (2022)